

ON EMBEDDING A GRAPH IN THE GRID WITH THE MINIMUM NUMBER OF BENDS*

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Abstract. Given a planar graph G together with a planar representation P , a region preserving grid embedding of G is a planar embedding of G in the rectilinear grid that has planar representation isomorphic to P . In this paper, an algorithm is presented that computes a region preserving grid embedding with the minimum number of bends in edges. This algorithm makes use of network flow techniques, and runs in time $O(n^2 \log n)$, where n is the number of vertices of the graph. Constrained versions of the problem are also considered, and most results are extended to k -gonal graphs, i.e., graphs whose edges are sequences of segments with slope multiple of $180/k$ degrees. Applications of the above results can be found in several areas: VLSI circuit layout, architectural design, communication by light or microwave, transportation problems, and automatic layout of graphlike diagrams.

Key words. planar embedding, grid, VLSI layout, bends, network flow

AMS(MOS) subject classifications. 68U05, 68R10, 68Q35, 94C15, 90B10

1. Introduction. Embedding a graph in the rectilinear grid has several applications. Among them we mention: VLSI circuit design [6], [12], architectural floor plan layout [7], [9], and aesthetic layout of diagrams used in Information Systems design [1], [11].

Given a planar graph G together with a planar representation P , a *region preserving grid embedding* of G is an embedding of G in the rectilinear grid such that

- (1) Paths of the grid that are mappings of edges are node-disjoint;
- (2) The planar representation of the embedding is isomorphic to P .

We consider the problem of minimizing the number of bends in a region-preserving grid embedding. Figure 1 shows an embedding of the complete 4-vertex graph that minimizes the bend count.

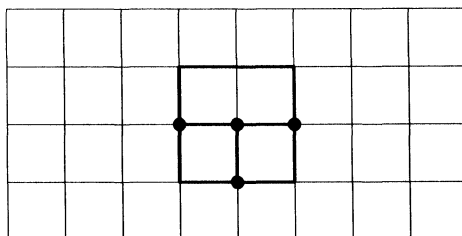


FIG. 1. A grid embedding with the minimum number of bends.

This problem has been explicitly considered in a paper of Storer [10], where the existence of a polynomial algorithm for its solution is left as an open problem. We settle this question here by showing an algorithm that finds an embedding with the fewest bends in $O(n^2 \log n)$ time, where n is the number of vertices of the graph. The basic idea of the algorithm is a transformation into a minimum cost flow problem.

The remaining sections of this paper are organized as follows. Section 2 contains basic definitions for a formal statement of the problem, and preliminary results. Section 3 is concerned with the transformation into min-cost-flow, and provides a worst case

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computing time analysis of the embedding algorithm. Section 4 considers constrained versions of the bend minimization problem. Section 5 shows an extension of the above transformation to “ k -angular” embeddings, where edges are mapped into sequences of segments with slope multiple of $180/k$ degrees.

2. Definitions and preliminary results. The basic definitions we adopt about graphs are essentially those in [3]. We allow graphs to have multiple edges and self-loops.

A *plane graph* $G = (V, E)$ is defined as follows: (1) V is a set of distinct points of the plane. (2) E is a set of simple curves connecting two points in V . (3) The simple curves in E do not cross each other, except, possibly, at endpoints. Elements in V are called *vertices*. Elements in E are called *edges*. A graph is *planar* if it is isomorphic to some plane graph. A plane graph divides the set of points of the plane that do not belong to edges into topologically connected regions, called *faces*. The unbounded region is referred to as the *external face*. If F is the set of faces of the plane graph $G = (V, E)$, *Euler’s formula* states that

$$|V| + |F| = |E| + 2.$$

The topological structure of a plane graph can be described by listing edges that appear on the contour of each face, and specifying the external face. We call *planar representation* of a plane graph G a set of circularly ordered edge lists $P(f)$, one for each face f . Edges in $P(f)$ appear in the same sequence as they are encountered when going around the contour of f in the “positive” direction, i.e., having the face at one’s right. Note that every edge of G appears exactly twice in the lists. If this happens in the same list $P(f)$, the edge is called *bridge*. An example of planar representation is shown in Fig. 2.

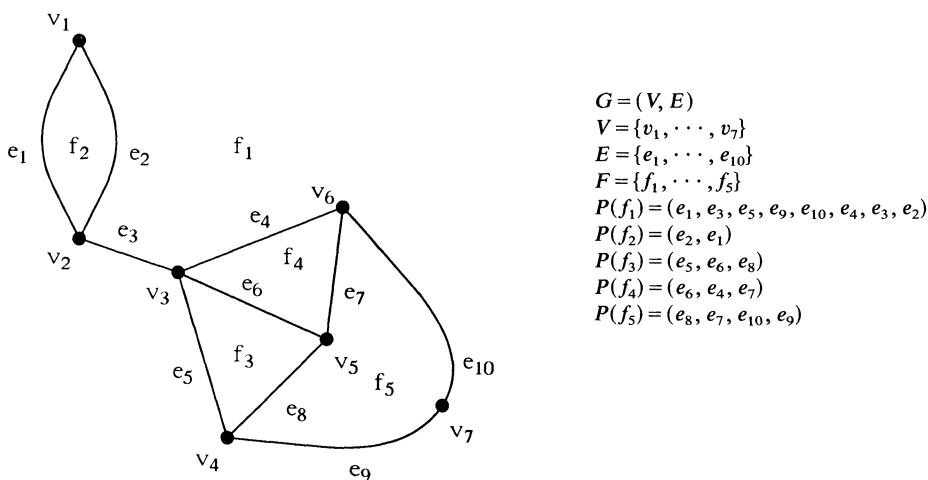


FIG. 2. Example of planar representation; edge e_3 is a bridge.

This definition can be extended to planar graphs as follows: let G be a planar graph; we say that P is a planar representation for G if P is the planar representation of some plane graph isomorphic to G .

The (*rectilinear*) *grid* is the infinite plane graph whose vertices have integer coordinates, and whose edges link pairs of vertices at unit distance. A (*planar*) *grid embedding* [12] of a graph G is a mapping Q of G into the grid such that

- (1) Q maps each vertex of G into a distinct point of the grid;

(2) Q maps edge e of G into a path of the grid $Q(e)$ whose endpoints are the mappings of vertices linked by e ;

(3) For every pair $\{e', e''\}$ of edges of G , the corresponding paths $Q(e')$ and $Q(e'')$ have no points in common, except, possibly, the endpoints.

The plane graph $Q(G)$ defined by the embedding Q is isomorphic to G . A graph admits a grid embedding if and only if it is planar and has maximum vertex degree less than or equal to 4 [12]. For short, we refer to these graphs as *4-planar* graphs.

In the following, we consider grid embeddings with a prespecified planar representation. Let $G = (V, E)$ be a planar graph, and P a planar representation for G . The grid embedding Q is called *region preserving* with respect to P if the plane graph $Q(G)$ has planar representation isomorphic to P [10].

We are now able to state formally the problem of our interest.

PROBLEM. *Grid embedding with the minimum number of bends.* Given a 4-planar graph G , and a planar representation P for it, find a grid embedding of G that is region preserving with respect to P and has the minimum number of bends.

Approximation algorithms for the above problem have been given by Storer [10]. Paoluzzi has developed a linear programming formulation for the case of biconnected graphs [9].

The concept of *orthogonal representation*, introduced below, will allow us to deal with the shape of a grid embedding, without taking into account dimensions (a similar concept has been developed by Paoluzzi [9] for biconnected graphs).

An *orthogonal graph* is a plane graph whose edges are alternating sequences of horizontal and vertical segments. Figure 3 shows an orthogonal graph.

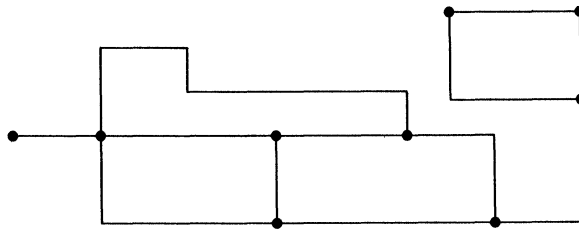


FIG. 3. Example of orthogonal graph.

The orthogonal representation describes the shape of an orthogonal graph, without considering segment lengths. This is obtained by enriching the lists of the planar representation with information about angles formed by edges. Let G be an orthogonal graph. The orthogonal representation of G is a set of circularly ordered lists $H(f)$, one for each face of G , whose elements have the format (e, s, a) , where

- (1) e is an edge of G ,
- (2) s is a binary string,
- (3) a is an integer in the set $\{90, 180, 270, 360\}$.

If r is an element of $H(f)$, its fields are denoted $e[r]$, $s[r]$, and $a[r]$. Each list $H(f)$ can be obtained from the corresponding list $P(f)$ of the planar representation by adding the s - and a -fields to each element, with the following meaning:

- (1) String $s[r]$ describes the shape of edge $e[r]$. The k th bit of $s[r]$ represents the k th bend that appears on the right side of $e[r]$, as it is encountered when going around f in the positive direction. The binary symbols **0** and **1** denote an angle of 90 and 270 degrees, respectively.

(2) Number $a[r]$ specifies the angle formed in face f by edges $e[r]$ and $e[r']$, where r' is the element following r in the list $H(f)$.

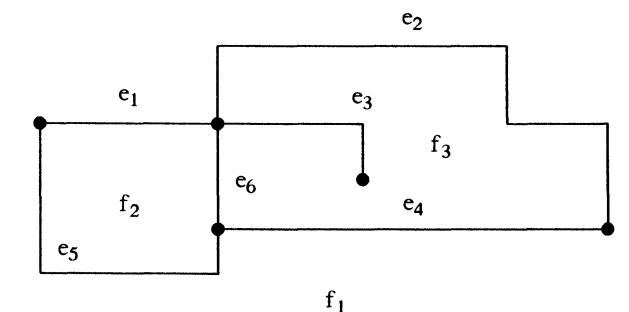
An example of orthogonal representation is given in Fig. 4.

The number of bends $B(H)$ of any orthogonal graph G having H as orthogonal representation is given by

$$(2.1) \quad B(H) = \frac{1}{2} \sum_f \sum_{r \in H(f)} |s[r]|$$

where $|s|$ is the length of string s .

For a set of circular lists H to be the orthogonal representation of some orthogonal graph, the following properties are necessary and sufficient, as can be proved by means of elementary geometric considerations.



$$H(f_1) = ((e_1, \varepsilon, 270), (e_5, 11, 90), (e_4, \varepsilon, 270), (e_2, 1011, 90))$$

$$H(f_2) = ((e_1, \varepsilon, 90), (e_6, \varepsilon, 180), (e_5, 00, 90))$$

$$H(f_3) = ((e_2, 0010, 90), (e_4, \varepsilon, 90), (e_6, \varepsilon, 90), (e_3, 0, 360), (e_3, 1, 90))$$

FIG. 4. Example of orthogonal representation. (The symbol ε denotes the empty string.)

PROPERTY 1. There is some 4-planar graph whose planar representation is given by the e -fields of the $H(f)$ lists.

PROPERTY 2. Let r' and r'' be a pair of elements in the orthogonal representation associated with the same edge. Then string $s[r']$ can be obtained from $s[r'']$ by reversing $s[r']$ and exchanging zeros with ones.

Property 2 means that each edge must have consistent descriptions in the faces in which it appears.

PROPERTY 3. Define, for each element r ,

$$\text{ROTATION}(r) = \text{ZEROS}(s[r]) - \text{ONES}(s[r]) + (2 - a[r]/90)$$

where $\text{ZEROS}(s)$ and $\text{ONES}(s)$ denote the number of zeros and ones in string s , respectively. The following equality holds:

$$\sum_{r \in H(f)} \text{ROTATION}(r) = \begin{cases} -4 & \text{if } f \text{ is the external face,} \\ +4 & \text{otherwise.} \end{cases}$$

Property 3 means that every face described by H is a rectilinear polygon, i.e. a polygon whose sides are horizontal and vertical segments.

PROPERTY 4. For each vertex v of the plane graph described by H , the sum of angles between pairs of consecutive edges around v , as given by the a -fields of the corresponding elements in the $H(f)$ lists, is equal to 360 degrees.

The orthogonal representation is dimensionless: it defines the reciprocal disposition of a set of horizontal and vertical segments. In order to construct from it a grid embedding, we have to compute the lengths of these segments. Three constraints must be satisfied:

- (1) All segment lengths are positive integers;
- (2) Every circuit of the graph is mapped into a rectilinear polygon;
- (3) No two segments intersect each other, except, possibly, at common endpoints.

An algorithm that accomplishes this task is outlined below.

ALGORITHM FIND_LENGTHS.

Input: Orthogonal representation H of a graph G .

Output: Grid embedding Q of G with orthogonal representation H .

Step 1. Construct a normalized orthogonal representation H' decomposing each face of H into rectangles with the introduction of fictitious edges and vertices.

Step 2. Assign consistently integer lengths to the sides of the rectangles in H' , in order to obtain a grid embedding Q' of the graph G' described by H' .

Step 3. Remove from Q' the fictitious vertices and edges introduced by Step 1. This gives the desired grid embedding Q .

Step 1 guarantees that no intersection will occur (constraint 3). It is a shape decomposition, since it operates at the orthogonal representation level. Step 1 can be carried out associating with each face f of H a circular binary string $S(f)$ given by

$$S(f) = s[r_0]w(a[r_0]) \cdots s[r_{n-1}]w(a[r_{n-1}])$$

where

$$H(f) = r_0, r_1, \dots, r_{n-1},$$

$$w(a) = \begin{cases} 0 & \text{if } a = 90, \\ \varepsilon & \text{if } a = 180, \\ 1 & \text{if } a = 270, \\ 11 & \text{if } a = 360. \end{cases}$$

The circular string $S(f)$ is searched for the pattern **100** and, if found, a rectangle is cut from f performing one of the two basic decomposition operations shown in Fig. 5. $S(f)$ is then updated substituting **0** for **100**, and the procedure is repeated until f becomes a rectangle. A slightly different treatment is given to the external face, which is decomposed into a set of rectangles plus the complement of a rectangle by means of a similar procedure.

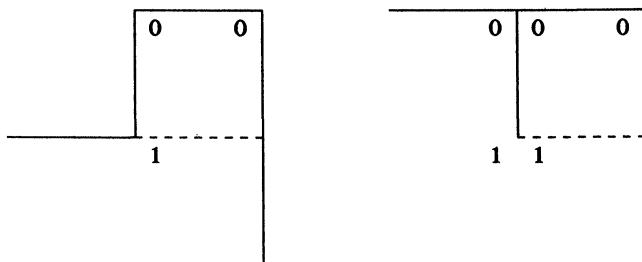


FIG. 5. Basic decomposition operations.

An efficient way to perform the above decomposition strategy is to stack the ones encountered while searching for **100**. This makes it possible to decompose each face f into at most $\text{ONES}(f) + 1$ rectangles in $O(|H(f)| + |S(f)|)$ time. Thus we have the following lemma.

LEMMA 1. *Step 1 of Algorithm FIND_LENGTHS, applied to an orthogonal representation H with n vertices and b bends, constructs a normalized orthogonal representation H' with $O(n + b)$ faces in $O(n + b)$ time.*

Step 2 can be efficiently performed using the compaction algorithm described in [4], whose complexity is linear in the number of faces of H' . Step 3 is easily carried out in time linear in the number of faces of H' .

THEOREM 1. *Algorithm FIND_LENGTHS embeds in the grid an orthogonal representation with n vertices and b bends in $O(n + b)$ time.*

Proof. The result follows immediately from Lemma 1 and the above discussion. Q.E.D.

In the rest of this section, we recall some definitions about network flow theory.

A network $N = (U, A, s, t, b, c)$ consists of

- (1) A finite set of nodes U ;
 - (2) Two distinguished nodes, s and t , called *source* and *sink*, respectively;
 - (3) A set A of ordered pairs of vertices, called *arcs*, that includes the special *return arc* (t, s) ;
 - (4) A positive *capacity* (bound) $b(a)$, for each arc a ;
 - (5) A *cost* $c(a)$, for each arc a .
- A *flow* in N is a function $x: A \rightarrow \mathbb{R}$ such that

- (1) $0 \leq x(a) \leq b(a)$ for every arc a , and
- (2) $\sum_u x(u, v) - \sum_w x(v, w) = 0$ for every node v

where each sum is over every node $u(w)$ for which the summand is defined.

The *value* of flow x , denoted $|x|$, is given by the flow in the return arc

$$|x| = x(t, s).$$

The *cost* of flow x is the quantity:

$$\text{COST}(x) = \sum_{a \in A} c(a)x(a).$$

The minimum cost flow problem asks for a flow of minimum cost among those with a given fixed value.

PROBLEM. MIN-COST-FLOW. Given a network N , and a positive number z , find a flow x in N of value $|x| = z$ that has minimum cost.

The *augmentation algorithm* for problem MIN-COST-FLOW consists of augmenting the initial zero flow along minimum cost paths [5]. In the case of integer capacities and nonnegative costs, the shortest path computations can be performed using Dijkstra's algorithm implementation with priority queues. Therefore, the augmentation algorithm has complexity $O(z(|U| + |A|) \log |U|)$.

A *flow-augmenting cycle* C with respect to flow x is a cycle in the undirected graph resulting from N by ignoring arc directions and removing the return arc (t, s) , with the following properties:

- (1) For every arc a traversed by C in the forward direction (called a *forward arc*), we have $x(a) < b(a)$;

(2) For every arc a traversed by C in the reverse direction (called a *backward arc*), we have $x(a) > 0$.

The *cost* of the flow-augmenting cycle C is given by the sum of the costs of the forward arcs minus the sum of the costs of the backward arcs.

3. Transformation of grid embedding into MIN-COST-FLOW. Let G be a 4-planar graph, and P a planar representation for G . In this section, we show that P can be extended to an orthogonal representation H with the minimum number of bends by computing a minimum cost flow in a network obtained from P through a straightforward transformation. A grid embedding for G can be constructed from the orthogonal representation H using the compaction algorithm described in the previous section.

The network $N(P) = (U, A, s, t, b, c)$ associated with the graph $G = (V, E)$ and the planar representation $P = \{P(f): f \in F\}$ is defined as follows:

Let $\hat{V}$ denote the subset of vertices of G having degree less than or equal to 3. Nodes of $N(P)$ are vertices in $\hat{V}$, faces of P , plus the source and the sink:

$$U = \hat{V} \cup F \cup \{s, t\}.$$

Arcs of $N(P)$ are of various types

$$A = A_V \cup A_F \cup A_S \cup A_T \cup \{(t, s)\}$$

where

(1) A_V contains arcs of the type (v, f) , where $v \in \hat{V}$ and $f \in F$, such that v is one endpoint of some edge in $P(f)$. Arcs in A_V have infinite capacity and zero cost. We define

$$E(v, f) = \{e \in P(f): \text{both } e \text{ and the next edge in list } P(f) \text{ are incident to vertex } v\}.$$

(2) A_F contains two arcs, (f, g) and (g, f) , for each pair $\{f, g\}$ of distinct faces such that the set

$$E(f, g) = \{e \in E: e \text{ appears in both } P(f) \text{ and } P(g)\}$$

is not empty. Moreover, A_F contains a self-loop (f, f) for each face f that has a bridge. In this case, we define

$$E(f, f) = \{e \in E: e \text{ is a bridge of face } f\}.$$

Arcs in A_F have infinite capacity and unit cost.

(3) A_S contains

(a) Arcs of the type (s, f) , such that f is an internal face whose list $P(f)$ has length $|P(f)| \leq 3$;

(b) Arcs of the type (s, v) , such that $v \in \hat{V}$.

Arcs in A_S have zero cost and capacity:

$$b(s, f) = 4 - |P(f)|, \quad b(s, v) = 4 - \deg(v).$$

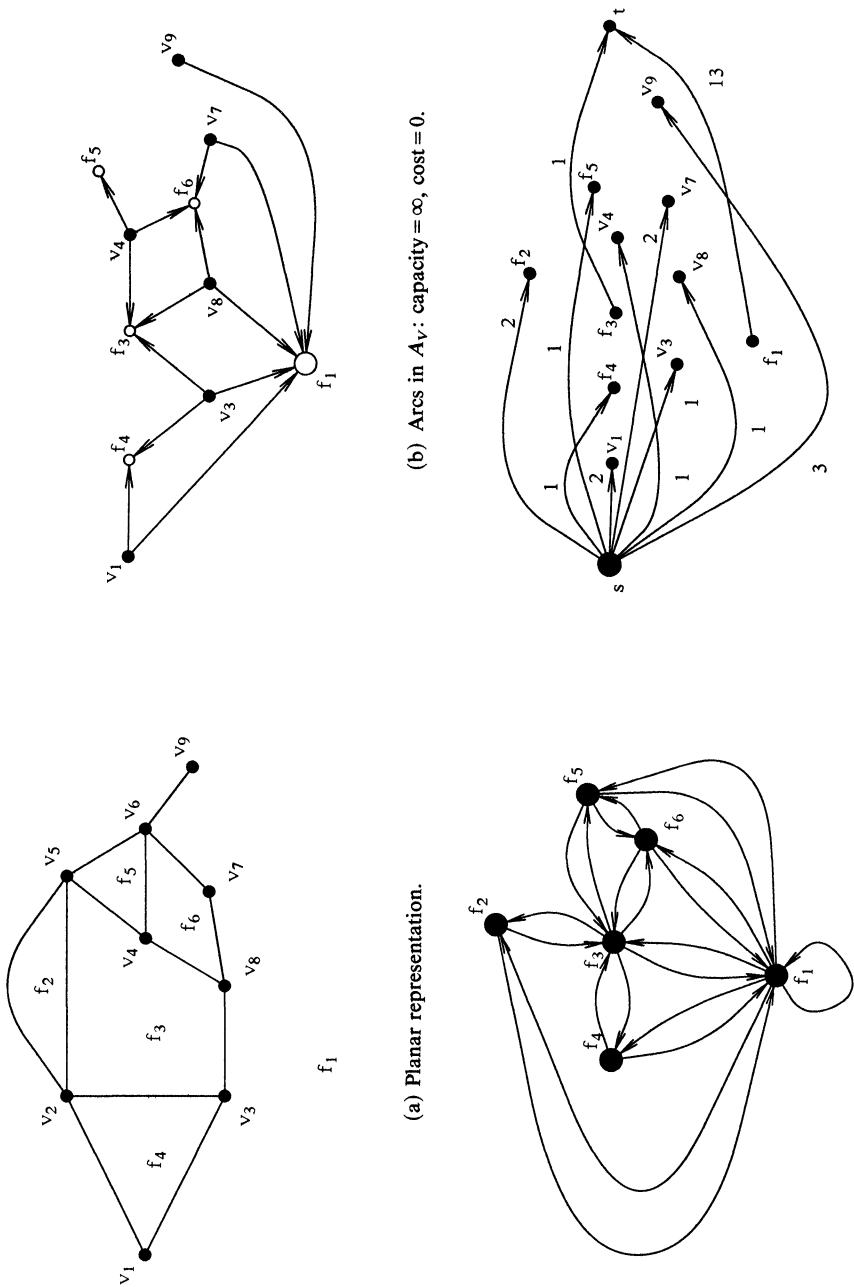
(4) A_T contains arcs of the type (f, t) , such that f is either the external face, or an internal face whose list $P(f)$ has length $|P(f)| \geq 5$. Arcs in A_T have zero cost and capacity given by

$$b(f, t) = \begin{cases} |P(f)| + 4 & \text{if } f \text{ is the external face,} \\ |P(f)| - 4 & \text{otherwise.} \end{cases}$$

See in Fig. 6 an example of this transformation.

The flow value in $N(P)$ is set equal to

$$z(P) = \sum_u b(s, u) = \sum_w b(w, t)$$



(a) Arcs in A_F : capacity = ∞ , cost = 1. (b) Arcs in A_V : capacity = ∞ , cost = 0. (c) Arcs in A_S and A_T : capacity is shown, cost = 0.

FIG 6. Example of network $N(P)$.

where the above equality follows from Euler's formula

$$\begin{aligned} \sum_{w \in U} b(w, t) - \sum_{u \in U} b(s, u) &= \sum_{f \in F} (|P(f)| - 4) + 8 + \sum_{v \in V} (\deg(v) - 4) \\ &= 2|E| - 4|F| + 8 + 2|E| - 4|V| \\ &= 4(|E| - |F| - |V| + 2) = 0. \end{aligned}$$

Because of the flow value $z(P)$, the flow in arcs of A_S and A_T must be equal to the capacity. Furthermore, since every node in $\hat{V}$ has exactly one ingoing arc with capacity at most 3, the flow in arcs of A_V cannot be greater than 3.

The transformation into a minimum cost flow problem is based on the following intuitive interpretation:

(1) Each unit of flow in network $N(P)$ represents an angle of 90 degrees: for arcs in A_V , flow $x(v, f)$ represents the sum of the measures of angles formed inside face f by the edges incident to v , which is given by $(x(v, f) + 1)90$; for arcs in A_F , flow $x(f, g)$ represents the number of bends that appear along the edges separating face f from face g and form an angle of 90 degrees inside face f .

(2) The conservation rule at vertex-nodes means that the sum of angles around each vertex must be equal to 360 degrees (Property 4). The conservation rule at face-nodes means that each face must be a rectilinear polygon (Property 3).

(3) The cost of the flow is equal to the number of bends.

LEMMA 2. *For each orthogonal graph G having planar representation P , there exists an integer flow x in $N(P)$ whose value is $z(P)$, and whose cost is equal to the number of bends in G .*

Proof. Let H be the orthogonal representation of G . The flow x in $N(P)$ is constructed from H as follows.

For each (v, f) in A_V , flow $x(v, f)$ is obtained from the a -fields of the elements of $H(f)$ associated with arc (v, f) :

$$x(v, f) = \sum_{r \in R(v, f)} (a[r]/90 - 1)$$

where

$$R(v, f) = \{r \in H(f) : \text{both } e[r] \text{ and the edge } e[r'] \text{ associated with the element } r' \text{ that follows } r \text{ in the list } H(f) \text{ are incident to vertex } v\}.$$

For each (f, g) in A_F , flow $x(f, g)$ is obtained from the s -fields of the elements of $H(f)$ associated with arc (f, g)

$$x(f, g) = \sum_{r \in R(f, g)} \text{ZEROS}(s[r])$$

where

$$R(f, g) = \{r \in H(f) : e[r] \in E(f, g)\}.$$

Flow $x(f, g)$ represents the angles of 90 degrees inside face f appearing along edges separating f from g . If $f = g$, then $x(f, f)$ is equal to the global number of bends along the bridges of f .

Flow in arcs of A_S and A_T is equal to the capacity, as imposed by the value of the flow. Figure 7 shows the above construction on the network of Fig. 6.

To prove that the function x so obtained is a legal flow in $N(P)$ we have to show that (1) capacity bounds are not violated; (2) flow is conserved at each node.

The first condition is easily verified from the above construction. The second condition is a consequence of the properties of the orthogonal representation H , as shown below.

For nodes in $\hat{V}$, we have from Property 4

$$\begin{aligned}\sum_f x(v, f) - x(s, v) &= \sum_f \sum_{r \in R(v, f)} (a[r]/90 - 1) - (4 - \deg(v)) \\ &= 360/90 - \deg(v) - 4 + \deg(v) = 0.\end{aligned}$$

For nodes in F , we have from Properties 2 and 3

$$\begin{aligned}\sum_u x(f, u) - \sum_w x(w, f) \\ &= \sum_g x(f, g) - \sum_g x(g, f) - \sum_v x(v, f) + |P(f)| \pm 4 \\ &= \sum_{r \in H(f)} (\text{ZEROS}(s[r]) - \text{ONES}(s[r]) - a[r]/90 + 1) + |P(f)| \pm 4 = 0.\end{aligned}$$

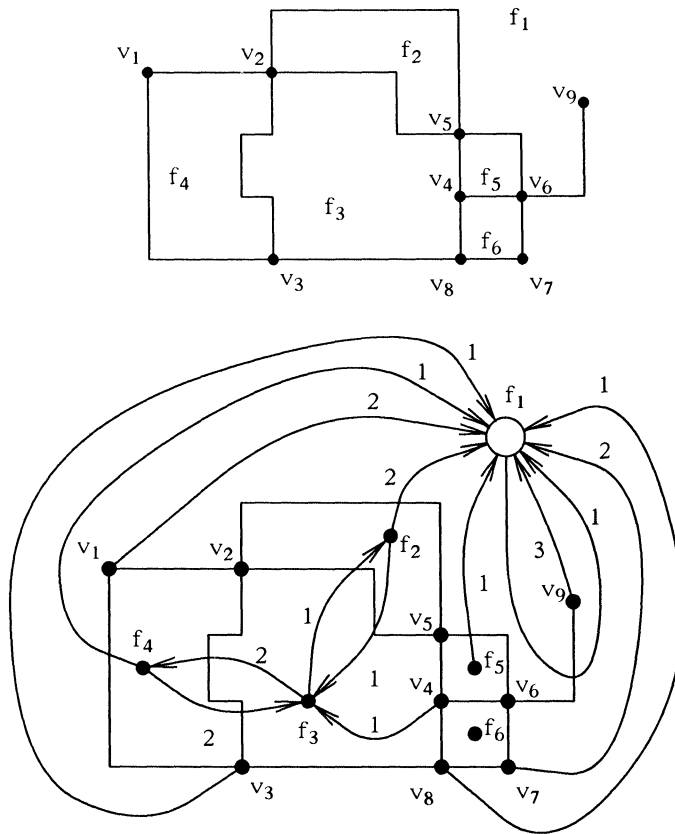


FIG. 7. Example of construction of flow x in network $N(P)$. (Only arcs in A_V and A_F with nonzero flow are shown.)

Finally, for nodes s and t , the conservation of flow derives immediately from the definition of the flow value $z(P)$.

To complete the proof observe that, from (2.1) and Property 2, the cost of flow x is given by

$$\begin{aligned} \text{COST}(x) &= \sum_{a \in A} c(a)x(a) = \sum_{(f,g) \in A_F} x(f,g) = \frac{1}{2} \sum_f \sum_{r \in H(f)} |s[r]| \\ &= B(H) = (\text{no. of bends in } G). \end{aligned} \quad \text{Q.E.D.}$$

LEMMA 3. *For each integer flow x in $N(P)$ of value $z(P)$, there exists an orthogonal graph G with planar representation P whose number of bends is equal to the cost of flow x . Furthermore, the orthogonal representation of G can be computed from flow x .*

Proof. We construct the orthogonal representation of G computing the a - and s -fields from the values taken by x in arcs of A_V and A_F , respectively.

For each arc (v, f) in A_V , let $r_1, \dots, r_n$ be the elements of $R(v, f)$ (see the definition of $R(v, f)$ in the proof of Lemma 2). Set

$$\begin{aligned} a[r_1] &= (x(v, f) + 1)90, \\ a[r_i] &= 90 \quad \text{for } 2 \leq i \leq n. \end{aligned}$$

For each pair of arcs (f, g) and (g, f) in A_F , let $r'_1, \dots, r'_n$ be the elements of $R(f, g)$, and $r''_1, \dots, r''_n$ be the corresponding elements of $R(g, f)$ (see the definition of $R(f, g)$ in the proof of Lemma 2). If $f = g$, then r'_i and r''_i are the pairs of elements in $H(f)$ corresponding to the two sides of the same bridge. Set

$$(3.1a) \quad s[r'_1] = \begin{cases} 0^{x(f,g)} & \text{if } f = g, \\ 0^{x(f,g)} 1^{x(g,f)} & \text{otherwise,} \end{cases}$$

$$(3.1b) \quad s[r''_1] = \begin{cases} 1^{x(g,f)} & \text{if } f = g, \\ 0^{x(g,f)} 1^{x(f,g)} & \text{otherwise,} \end{cases}$$

$$(3.1c) \quad s[r'_i] = s[r''_i] = \varepsilon \quad \text{for } 2 \leq i \leq n.$$

Property 1 is automatically satisfied, since we have built H from P . Property 2 easily follows from (3.1). Properties 3 and 4 follow from the conservation of flow in face-nodes and vertex-nodes, respectively. Finally, from (2.1) and (3.1) we have

$$\begin{aligned} B(H) &= \frac{1}{2} \sum_f \sum_{r \in H(f)} |s[r]| = \frac{1}{2} \sum_{(f,g) \in A_F} (x(f, g) + x(g, f)) \\ &= \sum_{a \in A} c(a)x(a) = \text{COST}(x). \end{aligned} \quad \text{Q.E.D.}$$

LEMMA 4. *There exists a flow of minimum cost in $N(P)$ that takes only integer values.*

Proof. Since the capacities in $N(P)$ are integers, the lemma follows from well-known results in network flow theory [5]. Q.E.D.

By putting together the previous lemmas, we obtain the following basic result.

THEOREM 2. *Let P be a planar representation. The minimum number of bends for any orthogonal graph with planar representation P is equal to the minimum cost of a feasible flow in $N(P)$ of value $z(P)$. Furthermore, the orthogonal representation H of any optimal orthogonal graph can be computed from some optimal flow in $N(P)$.*

The transformation into min-cost-flow provides a nice characterization of optimal orthogonal graphs.

LEMMA 5. *A flow x for network N is of minimum cost among all flows with the same value if and only if N has no flow-augmenting cycles (with respect to x) of negative cost.*

Proof. See [5].

THEOREM 3. *An orthogonal graph G has the minimum number of bends among all orthogonal graphs with the same planar representation if and only if there is not a directed closed simple curve J such that*

- (1) J intersects each edge of G at most twice;
- (2) J enters vertices of G only from angles greater than or equal to 180 degrees;
- (3) More than half of the edges crossed by J have a bend with the angle of 270 degrees on the side from which J enters.

Furthermore, if such a curve J exists, then an orthogonal graph G' can be found such that

- (1) G' has the same planar representation as G ;
- (2) G' has fewer bends than G ;
- (3) The orthogonal representation of G' differs from the one of G only in edges and vertices traversed by J .

Proof. We show that a curve J satisfying the above properties exists for G if and only if network $N(P)$ has a negative cost flow-augmenting cycle with respect to the flow x corresponding to the orthogonal representation H of G (see Lemma 2). The first claimed result then follows from Lemma 5.

Suppose that $N(P)$ has a negative cost flow-augmenting cycle $C = (a_1, \dots, a_m)$. C traverses neither the source s nor the sink t , since arcs incident to these nodes are saturated (see the definition of network $N(P)$). Construct from C a curve J , composed of stretches $J_1, \dots, J_m$, as follows:

- (1) Mark a point p_f in each face f of G ;
- (2) For each arc a_i of C , draw stretch J_i between the endpoints of a_i , i.e.
 - (i) if $a_i = (v, f) \in A_V$ is forward, then draw J_i from v to p_f inside face f ;
 - (ii) If $a_i = (v, f) \in A_V$ is backward, then draw J_i from p_f to v inside face f ;
 - (iii) If $a_i = (f, g)$ belongs to A_F , then draw J_i either from p_f to p_g , or from p_g to p_f , according to whether a_i is forward or backward. In the latter case, make J_i traverse an edge between f and g in correspondence to a bend, entering from the angle of 270 degrees.

Condition 1 is satisfied since J traverses edges of G only in correspondence to arcs of C in A_F , and each edge of G is associated with at most two arcs in A_F . Condition 2 of Theorem 3 is satisfied since, if J enters vertex v from face f , then (v, f) is a backward arc in C and has thus nonzero flow, corresponding to an angle of at least 180 degrees (see the proof of Lemma 2). Condition 3 follows from the fact that C has negative cost, and hence goes more times through backward arcs in A_F than it does through forward ones.

To prove the converse, suppose that a curve J satisfying conditions 1, 2, and 3 exists. First, decompose J into stretches $J_1, \dots, J_m$, where each J_i has one endpoint in a face, and the other endpoint either in a vertex or in a face; in the former case, J_i must not cross any edge, while, in the latter case, it must cross exactly one edge. Then, construct a flow augmenting circuit $C = (a_1, \dots, a_m)$ as follows:

- (1) If J_i goes from face f to vertex $v \in \hat{V}$, then set $a_i = (v, f)$; a_i is a backward arc of C since, from condition 2, $\text{arc}(v, f)$ has flow $x(v, f) \geq 1$;
- (2) If J_i goes from vertex $v \in \hat{V}$ to face f , then set $a_i = (v, f)$; a_i is a forward arc;
- (3) If J_i goes from face f to face g , then set $a_i = (g, f)$ backward if edge (f, g) has a bend with the angle of 270 degrees inside face f ; otherwise, set $a_i = (f, g)$ forward.

Conditions 1 and 2 assure that the circuit so constructed is a flow-augmenting cycle. Condition 3, together with the fact that arcs of type (v, f) have zero cost, assures that C has negative cost.

Finally, by augmenting the flow along cycle C (i.e. augmenting by a unit the flow in forward arcs, and decreasing by a unit the flow in backward arcs), we obtain a new flow x' with the same value as x , but with less cost. Lemma 3 implies then the existence of an orthogonal graph G' that satisfies the properties stated in the second part of the theorem. Q.E.D.

Figure 8 shows examples of application of Theorem 3, that can be used to prove the following intuitive facts.

COROLLARY 1. *Every bridge of an optimal orthogonal graph is a straight line segment.*

COROLLARY 2. *No edge of an optimal orthogonal graph has both a 90-degree and a 270-degree angle on the same side.*

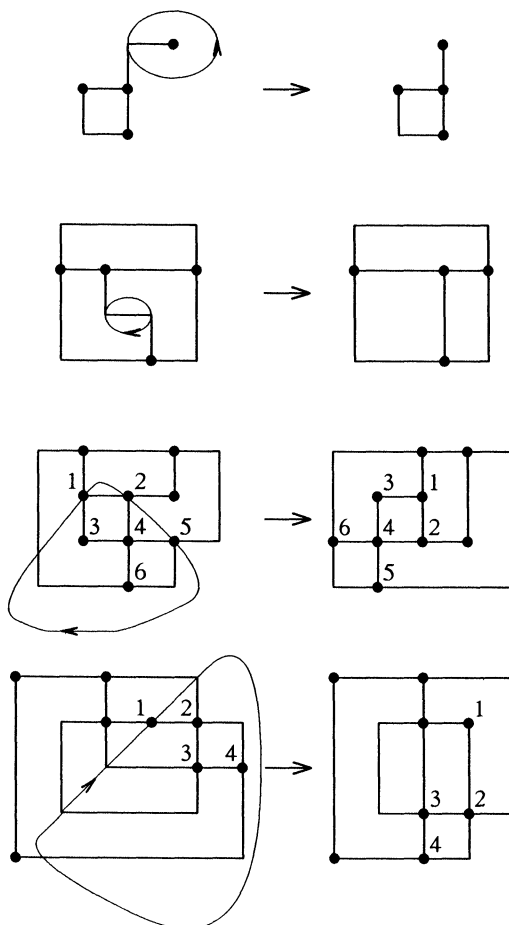


FIG. 8. Examples of application of Theorem 3.

We have shown that finding an orthogonal representation with minimum number of bends is equivalent to computing a minimum cost flow in network $N(P)$. The augmentation algorithm can be conveniently used for finding a minimum cost flow in network $N(P)$, since, as shown in the following, both the number of arcs and the value of flow in $N(P)$ grow linearly with the number of vertices of the graph represented by P .

LEMMA 6. *Let P be a planar representation for the 4-planar graph $G = (V, E)$. The following inequalities hold for the network $N(P) = (U, A, s, t, b, c)$ and the flow value $z(P)$:*

- (1) $|U| \leq \frac{3}{2}|V| + 4,$
- (2) $|A| \leq \frac{15}{2}|V| + 3,$
- (3) $z(P) \leq 3|V| + 2.$

Furthermore any one of them is satisfied as an equality for infinitely many planar representations.

Proof. We have from construction that

$$U = \hat{V} \cup F \cup \{s, t\}.$$

Let V_k be the subset of vertices with degree k . We have then

$$|U| = |E| - |V_4| + 4.$$

Since

$$(4.1) \quad 2|E| = \sum_k k|V_k|,$$

we obtain

$$|U| = \frac{1}{2}(|V_1| + |V_2| + 3|V_3| + 2|V_4|) + 4 \leq \frac{3}{2}|V| + 4$$

where the equality is attained when all vertices of G have degree 3. Figures 9(a) and 9(b) show examples of planar representations where this actually happens.

For the second inequality, we have

$$\begin{aligned} |A| &= |A_F| + |A_V| + |A_S| + |A_T| + 1 \\ &\leq 2|E| + 2|E| - 4|V_4| + |V| - |V_4| + |F| - |F_4| + 1 \end{aligned}$$

where

$$F_k = \{f \in F: f \text{ is internal and } |P(f)| = k\}.$$

Applying Euler's formula and (4.1) we obtain

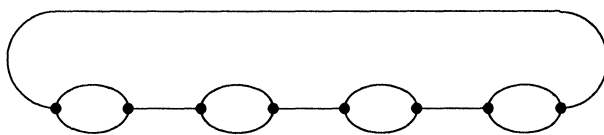
$$\begin{aligned} |A| &\leq 5|E| - 5|V_4| - |F_4| + 3 \\ &\leq \frac{5}{2}(|V_1| + 2|V_2| + 3|V_3| + 4|V_4|) - 5|V_4| - |F_4| + 3 \\ &\leq \frac{15}{2}|V| + 3. \end{aligned}$$

See in Fig. 9(b) an example of planar representation that satisfies formula 2 with equality.

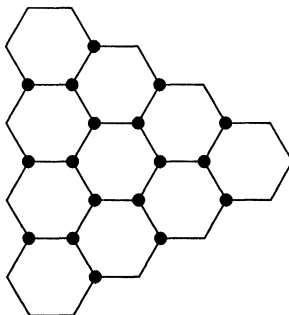
Finally, for the third inequality we have

$$z(P) = |P(q)| + 4 + \sum_{f \neq q} \max\{0, |P(f)| - 4\}$$

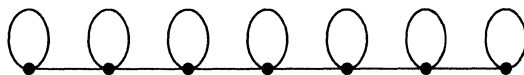
where q is the external face.



(a)



(b)



(c)

FIG. 9. Planar representations that attain the upper bounds of Lemma 6.

$$\begin{aligned}
 z(P) &= |P(q)| + \sum_{k=1}^{2|E|} |F_k|(k-4) - \sum_{k=1}^3 |F_k|(k-4) + 4 \\
 &= 2|E| - 4(|F| - 1) + \sum_{k=1}^3 |F_k|(4-k) + 4.
 \end{aligned}$$

We distinguish now two cases.

CASE 1. All vertices have degree 4.

At least one internal face has length greater than 1, since otherwise the graph would be a tree together with self-loops, and leaves would have odd degree. We have thus

$$z(P) \leq 4|V| - 4|F| + 3(|F| - 2) + 2 + 8 = 3|V| + 2.$$

CASE 2. Some vertex has degree less than four. Since in this case $|E| \leq 2|V| - 1$, we obtain

$$z(P) \leq 2(2|V| - 1) - 4|F| + 3(|F| - 1) + 8 = 3|V| + 2.$$

Planar representations like the one depicted in Fig. 9(c) attain the upper bound on $z(P)$. Q.E.D.

The following algorithm finds an orthogonal representation with the minimum number of bends.

ALGORITHM ORTHOGONALIZE.

Input. 4-planar graph G , planar representation P of G .

Output. Orthogonal representation H for P with the minimum number of bends.

Step 1. Construct network $N(P)$.

Step 2. Find a minimum cost flow x in $N(P)$, of value $z(P)$, by using the augmentation algorithm.

Step 3. Compute H from the optimal flow x , as stated in the proof of Lemma 3.

THEOREM 4. *Given a 4-planar graph G with n vertices, and a planar representation P for G , Algorithm ORTHOGONALIZE finds an orthogonal representation for P with the minimum number of bends in $O(n^2 \log n)$ time.*

Proof. Steps 1 and 3 of the algorithm are easily carried out in $O(|U| + |A|)$ time. Since in network $N(P)$ capacities are integer and costs are positive, Step 2 can be performed in $O(z(P)(|U| + |A|) \log |U|)$ time. The claimed result follows then from Lemma 6. Q.E.D.

If the graph G is not biconnected, it is convenient to apply Algorithm ORTHOGONALIZE separately to each biconnected component G_i of G . In order to guarantee a correct reconstruction of the global orthogonal representation, the capacity of each arc (v, f) in $N(P_i)$ associated with a cut-point v of G_i must be suitably decreased. The details are simple and left to the reader.

The complete algorithm for embedding a graph in the grid with the minimum number of bends consists of applying first ORTHOGONALIZE to find an optimal shape for the embedding, and then FIND_LENGTHS to give it dimensions.

ALGORITHM GRID_EMBED.

Input. 4-planar graph G , planar representation P for G .

Output. Grid embedding Q of G that is region preserving with respect to P , and has the minimum number of bends.

Step 1. Apply Algorithm ORTHOGONALIZE to find an optimal orthogonal representation H for P .

Step 2. Apply Algorithm FIND_LENGTHS to give lengths to segments in H , thus obtaining an optimal grid embedding Q of G .

Before stating the main theorem of the paper we need the following result of Storer.

LEMMA 7. *Every 4-planar representation with n vertices admits a grid embedding with no more than $3n + 2$ bends.*

Proof. See [10].

THEOREM 5. *Given a 4-planar graph G with n vertices, and a planar representation P for G , Algorithm GRID_EMBED finds a region preserving grid embedding of G with the minimum number of bends in $O(n^2 \log n)$ time.*

Proof. From Theorem 4, Step 1 of Algorithm GRID_EMBED has $O(n^2 \log n)$ time complexity. From Theorem 1, Step 2 has $O(n + b)$ complexity, where b is the number of bends in H . Since H is optimal, we have from Lemma 7 that $b = O(n)$. Therefore, the global time complexity of the embedding algorithm is $O(n^2 \log n)$. Q.E.D.

4. A constrained version of the bend minimization problem. The layout algorithm described in § 3 can be modified in order to take into account several types of constraints on the shape of the grid embedding, as well as piecewise linear convex cost functions for bends.

PROBLEM. *Shape constrained grid embedding with minimum bend cost.* Given:

- (1) A 4-planar graph $G = (V, E)$;
- (2) A planar representation P for G ;
- (3) A subgraph G' of G ;

- (4) An orthogonal representation H' for G' ;
- (5) An upper bound $U(e)$ on the number of bends, for each edge $e \in E'' = E - E'$;
- (6) A *bend cost* $d_e(k)$ for each edge $e \in E$, such that

$$d_e(0) = 0,$$

$$d_e(k+1) \geq d_e(k) \quad \text{for each } k \geq 0,$$

$$d_e(k+1) - d_e(k) \geq d_e(k) - d_e(k-1) \quad \text{for each } k \geq 1,$$

find a grid embedding Q of G subject to the following constraints:

- (1) Q is region preserving with respect to P ;
- (2) The orthogonal representation of the embedded subgraph $Q(G')$ is equal to H' ;
- (3) The number of bends $B(e)$ on each embedded edge $Q(e)$ verifies $B(e) \leq U(e)$;

that minimizes the global bend cost of Q , given by

$$\text{BEND_COST}(Q) = \sum_{e \in E} d_e(B(e)).$$

In order to deal with the new constraints, network $N(P)$ must be modified as follows.

Define $E''(f, g) = E(f, g) \cap E''$ and assign to each arc (f, g) in A_F the capacity

$$b(f, g) = \sum_{e \in E''(f, g)} U(e).$$

This guarantees that constraint (3) is satisfied in any optimal embedding (where $x(f, g)x(g, f) = 0$).

In order to satisfy constraint (2), each arc (f, g) of A_F such that the set $E(f, g)$ is contained in E' must have a flow of fixed value $x'(f, g)$, which can be computed from H' using the transformation described in the proof of Lemma 2. To obtain this, we replace arc (f, g) with two arcs, (s, g) and (f, t) , each having capacity equal to $x'(f, g)$. Moreover, for each arc (f, g) of A_F such that the set $E(f, g)$ has a nonempty intersection with E' but is not contained in it, we add the arcs (s, g) and (f, t) , but we do not remove the arc (f, g) . A similar transformation is performed on the arcs of A_V whose flow is imposed by H' .

As a result of this operation, parallel arcs might have been introduced in the network; in this case, we replace every bundle of parallel arcs $a_1, \dots, a_m$ with a unique arc a , summing up their capacities, i.e. $b(a) = b(a_1) + \dots + b(a_m)$. Further simplifications can now be performed, as explained below.

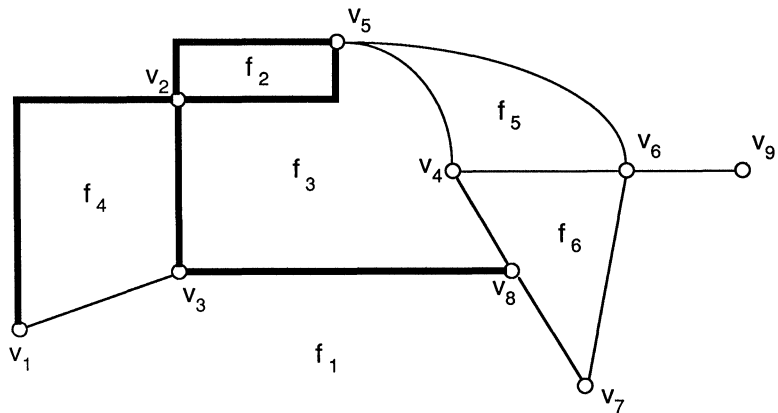
- (1) Remove nodes linked by arcs only to s and t ;
- (2) For each node u that is linked to both s and t
 - (i) if $b(s, u) = b(u, t)$, then remove both arcs (s, u) and (u, t) ;
 - (ii) if $b(s, u) > b(u, t)$, then remove arc (u, t) and reduce by $b(u, t)$ the capacity of arc (s, u) ;
 - (iii) if $b(s, u) < b(u, t)$, then remove arc (s, u) and reduce by $b(s, u)$ the capacity of arc (u, t) .

The flow value $z^*(P)$ in the resulting modified network $N^*(P)$ is set equal to

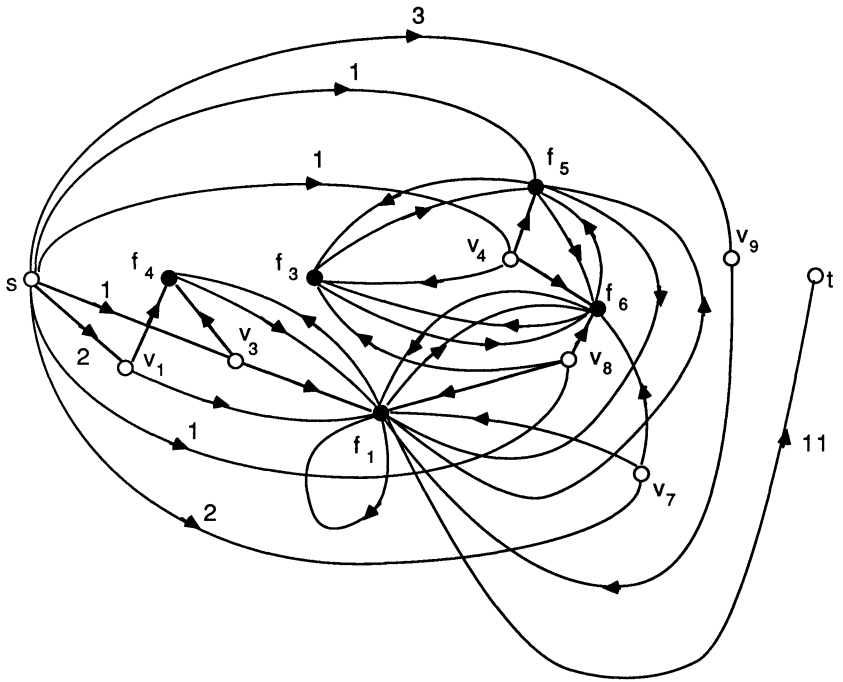
$$z^*(P) = \sum_u b(s, u) = \sum_w b(w, t).$$

Figure 10 shows an example of this construction.

A piecewise linear nondecreasing cost convex cost function $c_{(f, g)}(x)$ is associated with each arc (f, g) between face nodes, while the remaining arcs have zero cost. The cost function $c_{(f, g)}(x)$ is obtained from the bend costs of edges in $E''(f, g)$ as follows: Let $\Delta(f, g) = \delta_1, \dots, \delta_n, \dots$ be the list of values $d_e(k) - d_e(k-1)$ in nondecreasing



(a) Planar representation. Subgraph G' and its orthogonal representation are shown in bold lines.



(b) Network N^* : capacities are shown only for arcs incident to s or t .

FIG. 10. Example of construction of network $N^*(P)$.

order, where $e \in E''(f, g)$ and $1 \leq k \leq U(e)$; values are repeated in the list if they are generated by distinct edges e , or by distinct arguments k . For each arc (f, g) in A_F , the function $c_{(f,g)}(x)$ is given by

$$c_{(f,g)}(x) = \begin{cases} 0 & \text{if } x = 0, \\ \sum_{i=1}^k \delta_i & \text{if } x = k, k \in N, \\ c_{(f,g)}(k) + (x - k) \delta_{k+1} & \text{if } k < x < k + 1, k \in N. \end{cases}$$

The cost of a feasible flow x in network $N^*(P)$ is given by

$$\text{COST}(x) = \sum_a c_a(x(a)).$$

From the above discussion and the results in § 3, we obtain the following theorem.

THEOREM 6. *The minimum bend cost for any grid embedding Q that satisfies constraints (1), (2), and (3) is equal to the minimum cost of a feasible flow in $N^*(P)$ of value $z^*(P)$. Furthermore, the orthogonal representation H of any optimal grid embedding can be computed from some optimal flow in $N^*(P)$.*

Notice that the modified network $N^*(P)$ might not have any flow of value $z^*(P)$; this happens if and only if the constrained problem does not have any solution.

The out-of-kilter algorithm can handle piecewise linear convex cost functions without increase in the order of complexity [2], [5], [8]. This allows us to solve the constrained embedding problem by modifying Algorithm ORTHOGONALIZE of § 3 as follows.

ALGORITHM ORTHOGONALIZE2.

Input. See description of the problem *Shape constrained grid embedding with minimum bend cost*.

Output. Orthogonal representation H of a grid embedding Q that satisfies the constraints (1), (2), and (3), and has minimum bend cost.

Step 1. Construct network $N^*(P)$.

Step 2. If $N^*(P)$ does not have a feasible flow of value $z^*(P)$, report that the problem has no solution and stop. Otherwise, find a minimum cost flow $x^*(P)$ for $N^*(P)$ of value $z^*(P)$ by means of the out-of-kilter algorithm.

Step 3. Compute H from the optimal flow x^* and the orthogonal representation H' . Bends are assigned to edges of $E''(f, g)$ according to the sequence $\Delta(f, g)$, i.e. the number of bends in edge $e \in E''(f, g)$ is set equal to the number of occurrences of the differences δ_i computed from d_e in the sequence $\delta_1, \dots, \delta_{x(f,g)+x(g,f)}$.

5. Extension to k -gonal graphs. In this section, we show an extension of the previous results to a class of graphs that generalizes orthogonal graphs.

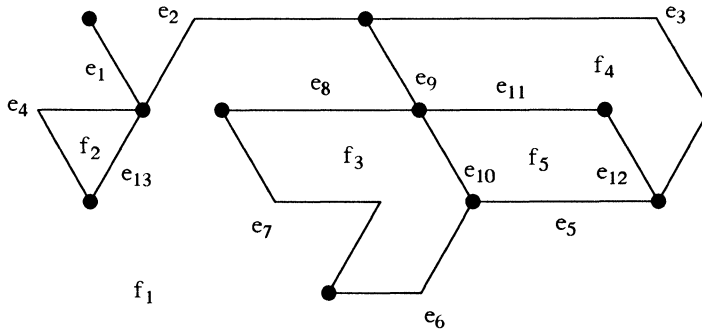
We call a k -gonal graph, where $k \geq 2$, a plane graph whose edges are sequences of segments with slope multiple of $180/k$ degrees, with respect to a reference axis. For $k = 2$, we obtain orthogonal graphs. The k -gonal representation describes the shape of a k -gonal graph G . It is a set of circularly ordered lists $K(f)$, one for each face f of the k -gonal graph, whose elements have the format (e, s, a) , where

(1) e is an edge of G ;

(2) s is a string from an alphabet of $2(k-1)$ symbols, which is useful to identify with the numbers $\pm 1, \dots, \pm(k-1)$;

(3) a is an integer in the set $\{m \cdot 180/k : m = 1, \dots, 2k\}$.

The meaning of the fields in an element r of $K(f)$ is the same as in the orthogonal representation, with the only difference that a symbol p in the string $s[r]$ stands for an angle of $180(1 - p/k)$ degrees. See the example in Fig. 11.



$$\begin{aligned}
 K(f_1) &= ((e_1, \varepsilon, 60), (e_4, +2, 300), (e_{13}, \varepsilon, 180), (e_2, +1, 120), \\
 &\quad (e_9, \varepsilon, 60), (e_8, \varepsilon, 300), (e_7, -1+2, 300), (e_6, -1, 120), \\
 &\quad (e_5, \varepsilon, 240), (e_3, -1-1, 180), (e_2, -1, 60), (e_1, \varepsilon, 360)) \\
 K(f_2) &= ((e_4, +2, 60), (e_{13}, \varepsilon, 60)) \\
 K(f_3) &= ((e_6, +1, 60), (e_7, -2+1, 60), (e_8, \varepsilon, 120), (e_{10}, \varepsilon, 120)) \\
 K(f_4) &= ((e_9, \varepsilon, 60), (e_3, +1+1, 60), (e_{12}, \varepsilon, 240), (e_{11}, \varepsilon, 120)) \\
 K(f_5) &= ((e_5, \varepsilon, 120), (e_{10}, \varepsilon, 60), (e_{11}, \varepsilon, 120), (e_{12}, \varepsilon, 60))
 \end{aligned}$$

FIG. 11. Examples of 3-gonal representation.

To each bend in a k -gonal graph a cost is given that increases with the deviation from the straight line, i.e., if a bend forms an angle α , its cost is $|\alpha - 180| \cdot k/180$. The *bend cost* of a k -gonal graph G is the sum of the costs of its bends and can be computed from its k -gonal representation K :

$$\text{BEND_COST}(G) = \frac{1}{2} \sum_f \sum_{r \in K(f)} \text{COST}(s[r]),$$

where, given $s[r] = p_1 \cdots p_n$, we define

$$\text{COST}(s[r]) = \sum_{i=1}^n |p_i|.$$

A necessary and sufficient condition for a set of circularly ordered lists $K(f)$ to be the k -gonal representation of some k -gonal graph is the satisfaction of the following four properties.

PROPERTY 1'. There is some $2k$ -planar graph whose planar representation is given by the e -fields of the $K(f)$ lists.

PROPERTY 2'. Let r' and r'' be a pair of elements in K associated with the same edge. Then string $s[r'']$ is obtained from $s[r']$ by reversing $s[r']$ and changing sign to all its symbols.

PROPERTY 3'. Define, for each element r

$$\text{ROTATION}(r) = \sum_{i=1}^n p_i + k(1 - a[r]/180)$$

where we denote $s[r] = p_1 \cdots p_n$. Then the following equality holds:

$$\sum_{r \in K(f)} \text{ROTATION}(r) = \begin{cases} -2k & \text{if } f \text{ is the external face,} \\ +2k & \text{otherwise.} \end{cases}$$

PROPERTY 4'. For each vertex v of the plane graph described by K , the sum of angles between pairs of consecutive edges around v , as given by the corresponding elements in the $K(f)$ lists, is equal to 360 degrees.

Given a $2k$ -planar representation P , the problem of finding a k -gonal representation K of minimum cost can be solved by computing a minimum cost flow in the network $N_k(P)$, which is defined in the same way as $N(P)$ (see § 3), but with the following changes:

- (1) $\hat{V}$ denotes the set of vertices with degree less than or equal to $2k-1$.
- (2) A_S contains an arc (s, v) for each vertex-node $v \in \hat{V}$. The capacity of these arcs is $b(s, v) = 2k - \deg(v)$.
- (3) A_S contains an arc (s, f) for each face f whose list $P(f)$ has length $|P(f)| < 2k/(k-1)$. The capacity of these arcs is $b(s, f) = 2k - |P(f)|(k-1)$.
- (4) A_T contains an arc (f, t) for each face f such that
 - (i) f has length $|P(f)| > 2k/(k-1)$, or
 - (ii) f is the external face.
 In case (i) we have $b(f, t) = |P(f)|(k-1) - 2k$, while in case (ii) we have $b(f, t) = |P(f)|(k-1) + 2k$.
- (5) The flow value in $N_k(P)$ is set equal to

$$z_k(P) = \sum_u b(s, u) = \sum_w b(w, t).$$

See the example in Fig. 12.

The following results are extensions of Lemmas 2-4 and 6, and of Theorems 2-4.

LEMMA 8. *For each k -gonal graph G having planar representation P , there exists an integer flow x in $N_k(P)$ whose value is $z_k(P)$, and whose cost is equal to the bend cost of G .*

LEMMA 9. *For each integer flow x in $N_k(P)$ of value $z_k(P)$, there exists a k -gonal graph G with planar representation P whose bend cost is equal to the cost of flow x . Furthermore, the k -gonal representation of G can be computed from flow x .*

LEMMA 10. *There exists a flow of minimum cost in $N_k(P)$ that takes only integer values.*

LEMMA 11. *Let P be a planar representation for the $2k$ -planar graph $G = (V, E)$. We have then, for the network $N_k(P) = (U, A, s, t, b, c)$ and the flow value $z_k(P)$,*

- (1) $|U| \leq (2k-1)|V|/2 + 4$,
- (2) $|A| = O(k|V|)$,
- (3) $z_k(P) = O(k|V|)$.

Furthermore, these bounds are attained in infinitely many planar representations.

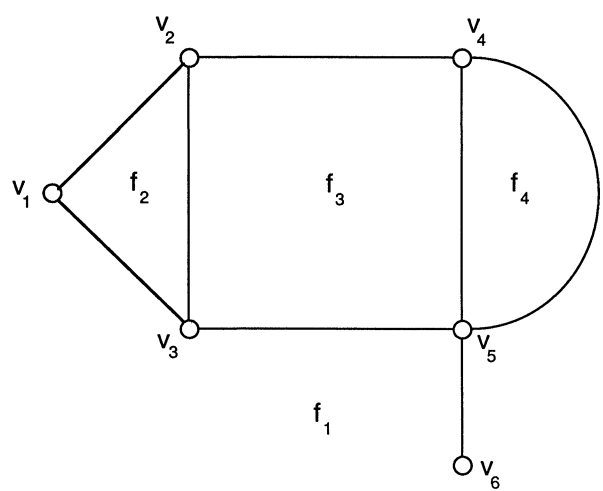
THEOREM 7. *Let P be a planar representation. The minimum bend cost for a k -gonal graph with planar representation P is equal to the minimum cost of a feasible flow in $N_k(P)$ of value $z_k(P)$. Furthermore, the k -gonal representation K of an any optimal k -gonal graph can be computed from some optimal flow in $N_k(P)$.*

THEOREM 8. *A k -gonal graph G has the minimum bend cost among all k -gonal graphs with the same planar representation if and only if there is not a directed closed simple curve J such that*

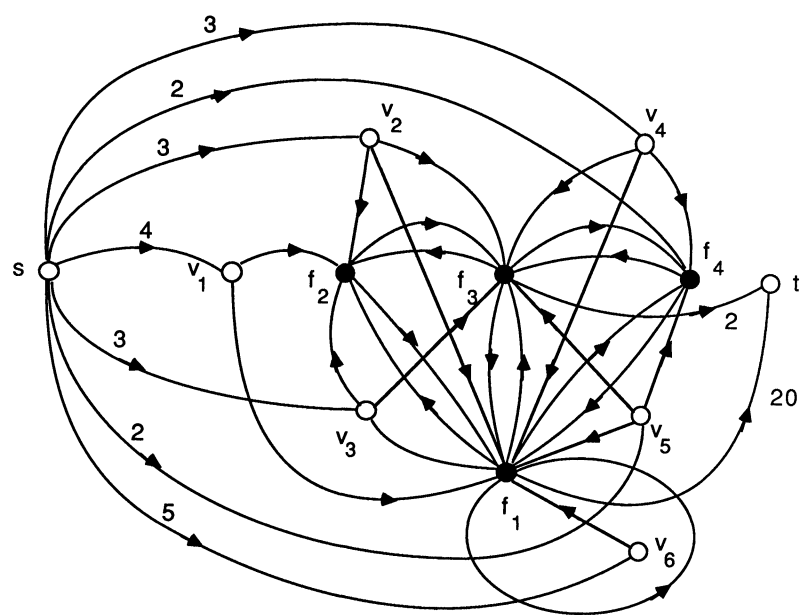
- (1) J intersects each edge of G at most twice;
- (2) J enters vertices of G only from angles greater than or equal to $360/k$ degrees;
- (3) More than half of the edges crossed by J have a bend that forms an angle greater than 180 degrees on the side on which J enters.

Furthermore, if such a curve J exists, then a k -gonal graph G' can be found such that

- (1) G' has the same planar representation as G ;
- (2) G' has bend cost less than the one of G ;
- (3) The k -gonal representation of G' differs from the one of G only in edges and vertices traversed by J .



(a) Planar representation P .



(b) Network $N_3(P)$: capacities are shown only for arcs in A_s and A_T .

FIG. 12. Example of network $N_3(P)$.

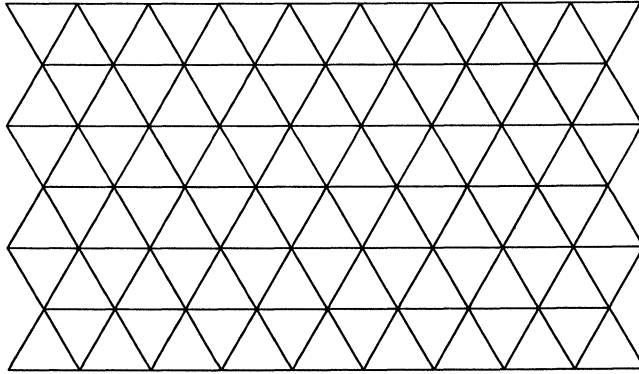


FIG. 13. Triangled grid for 3-gonal graphs.

THEOREM 9. *Given a $2k$ -planar graph G with n vertices, and a planar representation P for G , an orthogonal representation for P with minimum bend cost can be computed in $O(k^2 n^2 \log(kn))$ time.*

There is not a natural extension of the concept of grid to the general k -gonal case. For $k = 3$, the triangled grid shown in Fig. 13 seems to work. However, for $k > 3$, there is not an infinite plane graph with regular structure whose vertices all have degree $2k$ and whose edges are segments. This fact prevents us from giving an embedding algorithm for k -gonal representations similar to the one described in § 2.

Constrained versions of the bend cost minimization problem can also be considered, with results parallel to the ones presented in § 4.

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